
Infrastructure Restoration Planning

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Bayesian Linear Regression

§ 1.1

Basic Settings

Consider a training dataset

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N,$$

consisting of N observations, where $\mathbf{x}_i \in \mathbb{R}^p$ denotes the p -dimensional feature vector associated with the i -th observation and $y_i \in \mathbb{R}$ denotes the corresponding scalar response.

Define design matrix $\mathbf{X}$, target vector $\mathbf{y}$, and coefficient vector $\mathbf{w}$ as

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}_1^\top \\ \vdots \\ \mathbf{x}_N^\top \end{bmatrix} \in \mathbb{R}^{N \times p}, \quad \mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_N \end{bmatrix} \in \mathbb{R}^N, \quad \mathbf{w} = \begin{bmatrix} w_1 \\ \vdots \\ w_p \end{bmatrix} \in \mathbb{R}^p.$$

Under the linear regression model, each observation satisfies

$$y_i = \mathbf{w}^\top \mathbf{x}_i + \epsilon_i,$$

where the noise terms are assumed to be independent and identically distributed as

$$\epsilon_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2), \quad i = 1, \dots, N.$$

Define the noise vector

$$\boldsymbol{\epsilon} = \begin{bmatrix} \epsilon_1 \\ \vdots \\ \epsilon_N \end{bmatrix}.$$

It follows that

$$\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_N),$$

where $\mathbf{I}_N$ denotes the $N \times N$ identity matrix.



For the entire dataset, the linear regression model can therefore be written as

$$\mathbf{y} = \mathbf{X}\mathbf{w} + \boldsymbol{\epsilon}.$$

Consequently,

$$p(\mathbf{y}|\mathbf{X}, \mathbf{w}) \sim \mathcal{N}(\mathbf{X}\mathbf{w}, \sigma^2 \mathbf{I}_N).$$

Bayesian method includes two basic steps: inference and prediction, where inference is to estimate posterior $\mathbf{w}$, while prediction is to predict y^* given $\mathbf{x}^*$.

§ 1.2

Inference

1.2.1 Model Establishment

In Bayesian linear regression, the coefficient vector $\mathbf{w}$ is treated as a random variable and assigned a prior distribution. We assume that the prior distribution of $\mathbf{w}$ does not depend on $\mathbf{X}$, i.e., $w \perp X$, and thus $p(\mathbf{w} | \mathbf{X}) = p(\mathbf{w})$.

Given the training dataset, Bayesian formula is decomposed as:

$$p(\mathbf{w}|\mathcal{D}) = p(\mathbf{w}|\mathbf{X}, \mathbf{y}) = \frac{p(\mathbf{w}, \mathbf{X}, \mathbf{y})}{p(\mathbf{X}, \mathbf{y})} = \frac{p(\mathbf{w}, \mathbf{X}, \mathbf{y})}{p(\mathbf{y}|\mathbf{X})p(\mathbf{X})} = \frac{p(\mathbf{w}, \mathbf{y}|\mathbf{X})}{p(\mathbf{y}|\mathbf{X})} = \frac{p(\mathbf{y}|\mathbf{X}, \mathbf{w})p(\mathbf{w}|\mathbf{X})}{p(\mathbf{y}|\mathbf{X})} = \frac{p(\mathbf{y}|\mathbf{X}, \mathbf{w})p(\mathbf{w})}{p(\mathbf{y}|\mathbf{X})},$$

where the fifth equality follows from

$$p(\mathbf{y}|\mathbf{X}, \mathbf{w}) = \frac{p(\mathbf{w}, \mathbf{X}, \mathbf{y})}{p(\mathbf{w}, \mathbf{X})} = \frac{p(\mathbf{w}, \mathbf{X}, \mathbf{y})}{p(\mathbf{w}|\mathbf{X})p(\mathbf{X})} = \frac{p(\mathbf{w}, \mathbf{y}|\mathbf{X})}{p(\mathbf{w}|\mathbf{X})},$$

and the last equality follows from independency of prior coefficient and design matrix.

The denominator

$$p(\mathbf{y} | \mathbf{X}) = \int p(\mathbf{y} | \mathbf{X}, \mathbf{w})p(\mathbf{w})d\mathbf{w}$$

is the marginal likelihood. Since it does not depend on $\mathbf{w}$, the posterior distribution can be expressed up to a proportionality constant as

$$p(\mathbf{w}|\mathbf{X}, \mathbf{y}) \propto p(\mathbf{y}|\mathbf{X}, \mathbf{w})p(\mathbf{w}),$$

where $p(\mathbf{y}|\mathbf{X}, \mathbf{w})$ is a likelihood function and $p(\mathbf{w})$ is a prior function.

Suppose that a zero-mean Gaussian prior is imposed on the coefficient vector:

$$p(\mathbf{w}) = \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_p),$$



where $\Sigma_p \in \mathbb{R}^{p \times p}$ denotes the prior covariance matrix.

Since both the likelihood and the prior are Gaussian, their conjugacy implies that the posterior distribution of $\mathbf{w}$ is also Gaussian:

$$p(\mathbf{w}|\mathbf{X}, \mathbf{y}) = \mathcal{N}(\mu_w, \Sigma_w)$$

Thus,

$$p(\mathbf{w}|\mathbf{X}, \mathbf{y}) \propto \mathcal{N}(\mathbf{X}\mathbf{w}, \sigma^2 \mathbf{I}_N) \mathcal{N}(\mathbf{0}, \Sigma_p).$$

The remaining task is therefore to derive the posterior mean μ_w and posterior covariance Σ_w .

1.2.2 Model Solution

We first write $\mathcal{N}(\mathbf{X}\mathbf{w}, \sigma^2 \mathbf{I}_N) \mathcal{N}(\mathbf{0}, \Sigma_p)$ in explicit form:

$$\mathcal{N}(\mathbf{X}\mathbf{w}, \sigma^2 \mathbf{I}_N) \mathcal{N}(\mathbf{0}, \Sigma_p) = \frac{1}{(2\pi)^{N/2} \sigma^N} \exp\left(-\frac{1}{2}\right)$$

§ 1.3

Prediction